

JSGP · SHINE HACKATHON

Morning Triage

The morning triage for One Care caseworkers —
who needs you first, and why, in plain language.

THE PROBLEM

The safety net is **reactive**

Alarm buttons and wearables fail exactly when they're needed — an unconscious senior presses nothing.

A fall becomes a **long lie**: hours helpless on the floor.

THE INSIGHT

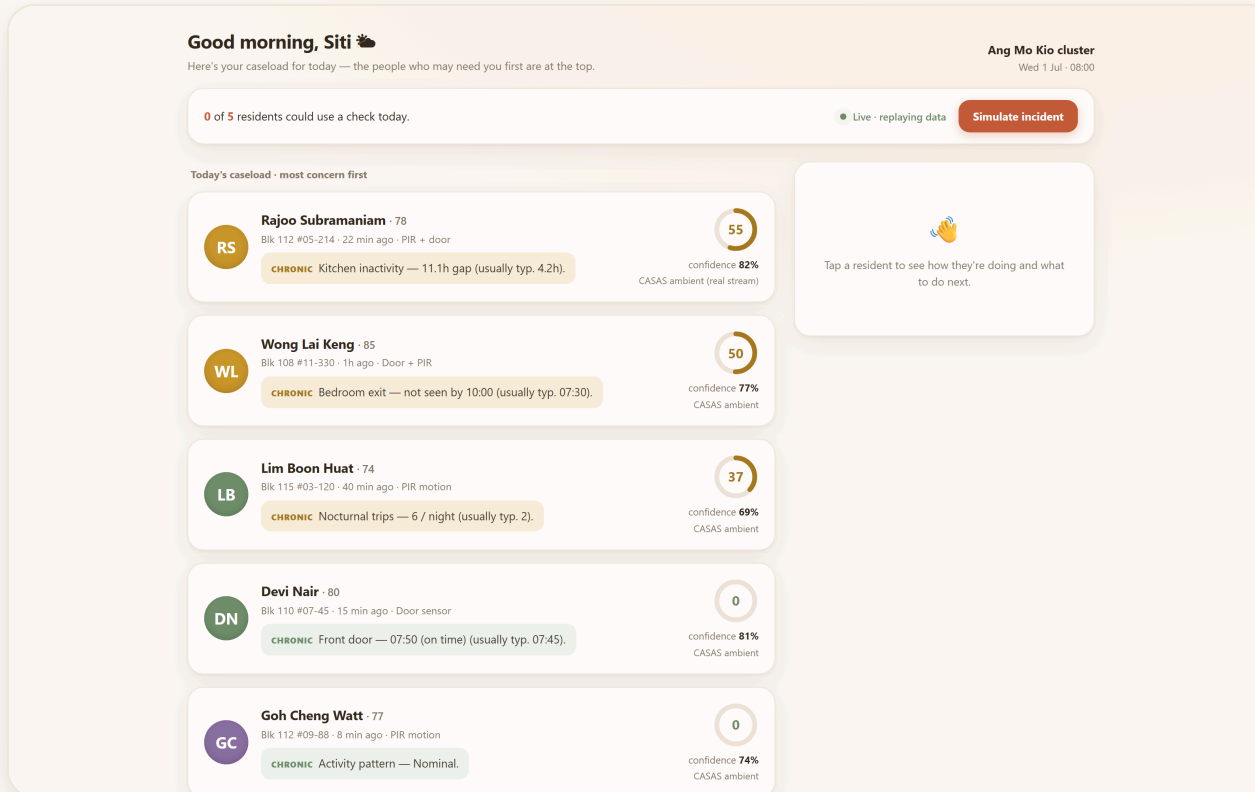
The home **already** knows

Routine is a vital sign. Absence of motion is data.

The senior presses nothing, wears nothing new,
and is never on camera.

THE PRODUCT

One screen, calm by default



Every resident ranked by need — each row states its reason in one sentence.

THE DEMO BEAT

A fall preempts everything

0 of 5 residents could use a check today.

● Live · replaying data

Simulate incident

One button stands in for a real fall signal

FALL DETECTED

TA

Tan Ah Moi · 82
Blk 108 #04-210 · just now · Accelerometer

ACUTE Fall detected — 11.2 g impact then 12 s no movement.

100

confidence **100%**
Wearable bangle

Call now

Escalate

A fall preempts everything — seconds, not a morning

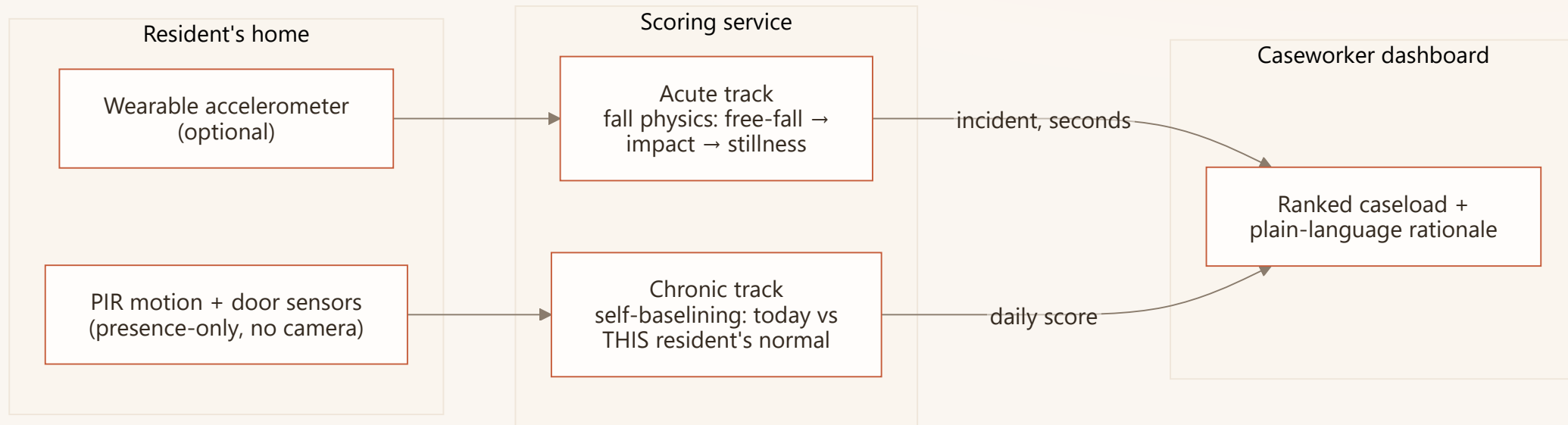
WHAT TO DO NEXT

CALL NOW. If no response, escalate to MOH 24/7 line + dispatch.

The system suggests; the caseworker acts

HOW IT WORKS

Sense → Score → Decide



Deterministic, explainable heuristics — no black box between sensor and caseworker.

EXPLAINABLE BY CONTRACT

Every number **decomposes**

WHY SHE'S RANKED HERE

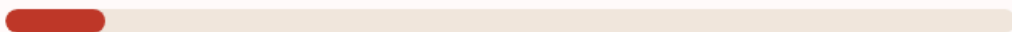
Peak impact **11.2 g**
walking < 1.4 g



Post-impact stillness **12 s**



Orientation **Horizontal**



Every score decomposes — nothing to take on faith

100
ACUTE RISK

100%
CONFIDENCE

Accelerometer
SOURCE

Trust is shown separately from risk
The system ranks and explains.
A human decides.

VALIDATED ON REAL DATA

It's **real**, not a concept

96.2%

FALLS DETECTED · 1,798 REAL FALLS

0–5%

FALSE ALARMS · ELDERLY-TYPICAL

220 days

REAL HOME · CHRONIC TRACK



Rajoo Subramaniam · 78

Blk 112 #05-214 · 22 min ago · PIR + door

CHRONIC Kitchen inactivity — 11.1h gap (usually typ. 4.2h).



confidence **82%**
CASAS ambient (real stream)

Rank 1: a real 220-day routine, broken — in plain words

BUILT FOR SINGAPORE

No imported **prior**

- **Self-baselining** — learns each resident's own rhythm in ~2 weeks; no foreign routine assumed.
- **Presence-only dignity** — seniors accept what doesn't watch them.
- Deploys **per HDB unit**; multiplies One Care's existing people, replaces no one.

HONEST LIMITS

What we **don't** claim

- Shrinks time-to-detection — does **not** promise to catch every event.
- Fall thresholds calibrated on mostly-young-adult recordings; the senior false-alarm rate is extrapolated.
- Irregular lives get wide baselines — surfaced as **low confidence**, never hidden.
- ~2 weeks of history before chronic scores mature.

BEYOND THE POC

From demo to **pilot**

- **Escalation integrations** — welfare-call automation, MOH/AIC line handoff on unacknowledged falls.
- **Caseworker briefing** — auto-drafted, grounded strictly in the deterministic features.
- **Pilot design** — one One Care cluster; measure time-to-detection and caseworker minutes saved.

THE ASK

Pilot **with us**

One cluster. Commodity sensors. A working system —
validated on **4,505 real recordings** and a **220-day real home**.

Built at SHINE Hackathon · demo runs live.